

PULSE: Payment Understanding, Lifecycle Scoring & Embeddings

A Transaction Foundation Model on Synthetic ISO 20022 Rails - a Falsifiable Replication and Extension of arXiv:2410.07851

Subrato B. · subrato.b@outlook.com

Preprint · July 2026 · code and generators: github.com/Bratz/txn-repr-poc

Abstract. We present PULSE, a transaction foundation model built by replicating the architecture of Raman et al. (arXiv:2410.07851) on synthetic ISO 20022 payments and measuring, against thresholds fixed before each run, what a single frozen tabular encoder can and cannot predict about a payment. Five claims were pre-registered. Two replicate the source paper: the partitioned high-cardinality embedder matches dense masked-column reconstruction at 5.8% of the embedding parameters (C1), and the frozen-encoder-plus-adapter stack trains 0.59% of a full fine-tune's parameters but loses to CatBoost by 44.5 PR-AUC points on a rule-generated risk label (C2) - trees read rule labels directly; the encoder bottleneck cannot. Three claims concern sequences. A small history encoder over frozen per-row embeddings beats an order-blind pooled baseline by 38.5 PR-AUC points (C3) and CatBoost on hand-built aggregates by 41.3 points (C4) on a regime-change task whose order-invariant statistics are matched across classes. Feeding the same history representation through a frozen Phi-1.5 with 7.6M trainable adapter parameters buys 1.3 points over a linear probe (C5) - under our 2-point bar, so serving drops the LLM. We then extend the corpus to a nine-message ISO 20022 lifecycle (pain.001 through camt.056/pacs.004) and report, wins and nulls alike: settlement-time regression beats its naive baseline at initiation (MAE 172 vs 234 minutes), in-flight outcome scoring sharpens from 0.86 to 0.94 PR-AUC as messages arrive, and fusing pain.001 origination context lifts account-takeover detection from chance (0.03) to 1.00; message-order signal is a measured null (-0.5 points), and cancel/return heads sit at prevalence. All data is synthetic and seeded; 197 tests reproduce every number.

1 Introduction

A payments engine asks many questions about the same row. Which rail should carry this transfer? Will it settle straight through, and if not, which exception fires? When does the beneficiary see funds? Is this account suddenly bursting with activity it never showed before? Production systems typically answer each question with its own model, its own features, and its own retraining calendar. Raman et al. [1] propose the alternative this paper tests: pretrain one tabular encoder on the payment corpus, freeze it, and answer every downstream question with a small adapter or probe on the frozen representation.

Replications in this space tend to fail quietly - a retuned learning rate here, a relabeled dataset there, and the headline number survives while the claim it supported dies. We took the opposite discipline. Every architectural choice is pinned to the source paper's section that mandates it, thresholds live in a config file committed before the runs, and a miss stays a miss: no post-hoc retuning closed any gap reported here. The result is a ledger rather than a victory lap. Some claims replicate strongly. One fails against our own bar, and the failure is informative. One resolves a design fork - whether the frozen LLM earns its inference cost - with a number instead of an opinion.

Our contributions: (i) an independent replication of the source paper's parameter-efficiency claims at near-paper vocabulary scale (119,819 accounts) on fully synthetic data; (ii) three pre-registered sequence claims measured on a corpus whose order-invariant statistics are matched across classes, isolating temporal signal by construction; (iii) a measured answer to the frozen-LLM question - 1.3 PR-AUC points for 7.6M adapter parameters plus a 1.3B-parameter forward pass; (iv) an extension of the single-table setting to a nine-message ISO 20022 lifecycle with event-anchored timestamps, recall and return legs, and a second data source

(pain.001 origination context) whose fusion we evaluate; and (v) a capability-routing discipline that names which questions the model refuses to answer because a rule, a lookup, or a template answers them exactly.

2 Related work

Transaction representation learning has moved from sequence models over card streams - TabBERT [7], CoLES [8], FATA-Trans [9] - toward foundation-model framings in which one pretrained backbone serves many tasks: TransactionGPT [10] at Visa and PRAGMA [11] at Revolut are the production-scale statements of that position. Raman et al. [1], our source, contribute the tabular side: columns as tokens, a partitioned embedder for high-cardinality identifiers, an offline party encoder, and a frozen LLM decoder adapted with small trainable modules in the prefix-tuning family [4]. The C2 finding we report - gradient-boosted trees [6] winning on labels that are rules over raw features - repeats a decade of tabular-benchmark results and is why we treat CatBoost as the bar every claim must clear, not as a strawman. TabPFN [12] offers a different foundation-model bet (in-context learning over small tables); we assessed it as out of scope because our tables exceed its row and class budgets and our deployment needs a persistent, incrementally probed representation.

3 Synthetic corpora

No real transaction data appears anywhere in this work. Three generators, all seeded and committed, produce the corpora; identical commands reproduce byte-identical files on any machine.

3.1 pacs.008 corpus (claims C1, C2)

One million interbank credit transfers over 19,895 corporate parents and a combined account vocabulary of 119,819 - close to the source paper's ~125K. The risk label is deliberately a transparent rule over raw fields (cross-border flag, currency, region, industry, amount bands), with the High class at 1.8% prevalence. That choice is honest hostility: a label a tree can read directly is the hardest case for an encoder that must compress the row into one vector before any head sees it.

3.2 Behavioural corpus (claims C3-C5)

Per-account payment histories, 8 to 40 events each, where 45% of accounts undergo a regime change midway through their stream. The generator matches the gap multiset and the amount distribution across the two classes, so every order-invariant aggregate - means, quantiles, counts, totals - carries no class information. Only a model that reads the ordered, timed sequence can separate the classes. This construction is what lets us attribute C3 and C4 to temporal signal rather than to feature leakage. The full corpus yields 24,016 sequences; evaluation holds out 20% of accounts (4,803 sequences, positive prevalence 0.450), never rows within a training account.

3.3 India-rails lifecycle corpus (Section 6)

Twenty thousand payments across RTGS, NEFT, IMPS, and SWIFT, each expanded into its ISO 2002 message trail: pain.001 acceptance, pain.002 status, pacs.008 clearing (pacs.009 cover where correspondent banking requires it), pacs.002 as the gpi tracker backbone, camt.054 credit notification, camt.056/camt.029 recall legs, and pacs.004 returns. Message timestamps anchor to generated exception events rather than to interpolation, and each message carries a direction and a visibility flag - an inward payment's pain.* legs live at the remote bank and never enter our prefixes, which is the information set a real engine has. A separate origination table attaches channel, device, session, and authentication attributes to each outward pain.001 at t=0 (10,943 rows), simulating the data a bank's own channel layer possesses before any ISO message exists. A cadence-enabled variant of the same generator (salary, rent, utility, supplier schedules) supports the forecasting heads.

4 Architecture

The pinned encoder follows the source paper: each column embeds as one token (25M parameters total), high-cardinality identifiers route through the partitioned embedder of their Section 3.1, party identities come from an offline-pretrained persistent store, and amounts pass through a currency-conditioned adaptive quantizer. Pretraining combines masked-column reconstruction with a batch-hard triplet objective [2], three epochs, then the encoder freezes for good. Nothing downstream ever updates it.

For sequences, a small transformer (the history encoder) consumes the frozen per-row embeddings of an account's ordered payments together with time deltas and calendar features, pretrained by reconstructing held-out field values, and emits one history vector per prefix. Downstream heads then take one of two forms. Option A is a linear probe on the history vector. Option B routes the same vector into a frozen Phi-1.5 [5] through trainable adapters in the prefix family [3, 4] - 7,638,528 trainable parameters against the LLM's 1.3B frozen ones. C5 is the measured comparison between the two.

Not every question reaches the model. Rail caps and floors, assigned reference values, identifier lookups, and duplicate checks are answered by guards and templates that are exact by construction; we delisted those tasks from the model's advertised capabilities after finding early prototypes claiming credit for predictions a two-line rule makes with certainty. The serving layer routes each field to model, rule, lookup, or template, and the documentation names which is which. A model that answers everything answers some things worse than the infrastructure it runs on.

5 Pre-registered claims

Table 1 states each claim, its threshold as committed in configuration before the run, and the measured outcome. Full-scale numbers come from single seeded runs on one NVIDIA H200; smoke-scale runs on CPU reproduce every qualitative verdict.

Claim	Measured	Threshold	Verdict
C1 partitioned embedder: parameter ratio vs dense, at matched reconstruction	0.058 ; recon gap -0.012 pp (partitioned marginally better)	ratio $\leq$ 0.55; gap $\leq$ 1 pp	pass
C2a adapter parameter efficiency	0.0059 of full fine-tune (7.64M / 1.3B)	$\leq$ 0.10	pass
C2b adapter beats CatBoost on rule-label risk	PR-AUC 0.210 vs 0.655 (-44.5 pp)	$\geq$ +10 pp	fail
C3 temporal lift: history encoder vs order-blind pooling	0.919 vs 0.533 (+38.5 pp)	$\geq$ +10 pp	pass
C4 held-out generalisation vs CatBoost on aggregates	0.919 vs 0.506 (+41.3 pp)	$\geq$ +5 pp	pass
C5 LLM necessity: frozen Phi-1.5 + adapters vs linear probe	0.932 vs 0.919 (+1.3 pp for the LLM)	drop the LLM if within 2 pp	drop the LLM

Table 1: the five claims. C1/C2 on the 1M-row pacs.008 corpus (positive class 1.8%); C3-C5 on the behavioural corpus, held-out accounts, PR-AUC on regime change. Thresholds predate the runs; no configuration was retuned after a miss.

5.1 What replicated

C1 is the source paper's core engineering claim and it replicates cleanly: the partitioned embedder holds reconstruction quality while spending 5.8% of the dense embedding budget, at a vocabulary within a few percent of the paper's. C3 and C4 are the strongest results in this report. On data where every order-invariant aggregate is uninformative by construction, CatBoost sits near chance (0.506) while the history encoder reaches 0.919 on unseen accounts - the first configuration in this project where the learned representation beats the tree, and it does so because the signal is genuinely temporal. A companion velocity task (does this account's next window contain an activity burst, prevalence 0.131) shows the same shape: time-aware 0.822 against order-blind 0.222.

5.2 What failed, and why we kept it

C2b failed against our own bar. The adapter learns real signal - 0.210 PR-AUC is roughly ten times the prevalence floor - but CatBoost reads the label-generating rule straight off the raw features, and no amount of frozen-representation elegance competes with direct access to the rule. We could have made this number respectable by softening the label. We left it, because it draws the deployment boundary honestly: on complete rows with rule-like labels, ship the tree. The encoder earns its keep where trees cannot follow - order, timing, partial information, and fusion across sources - which is precisely where Sections 5.1 and 6 place it.

5.3 The LLM verdict

C5 asks whether the frozen LLM belongs in the serving path. On regime classification from the same frozen history vector, Option B (Phi-1.5 plus adapters) reaches 0.932 against the probe's 0.919. The gap is real and it is small: 1.3 points, under the 2-point bar we set in advance, purchased with 7.6M trained parameters and a 1.3B-parameter forward pass per prediction. An earlier mock-LLM control had suggested the LLM subtracts 13 points; the real model corrects the sign but not the decision. Serving therefore uses linear probes and isotonic calibration on the frozen representations, and nothing in the deployed path executes an LLM. We note the untested cases plainly: tasks that emit text, few-shot task switching, and instruction-conditioned multi-task heads are the settings where the LLM could still earn its cost, and none of them is measured here.

6 The lifecycle extension: wins and nulls

The India-rails corpus turns the single-table problem into an engine's problem: score at intake, rescore as each message lands, and answer questions no single message contains. Table 2 reports every prediction the system exposes, including the ones that lose. Splits hold out accounts; in-flight prefixes exclude the messages that state the outcome being predicted.

Prediction	Measured	Baseline	Reading
Settlement ETA at initiation (pain.001 only)	MAE 172 min	234 (naive)	win, pre-clearing
Settlement ETA at intake (full row, n=4,000 eval)	MAE 195 min	392 (naive)	win
Charges	MAE 117	203 (naive)	win
'Will it credit?' as messages arrive	PR-AUC 0.860 -> 0.942 (prefix 1 -> 3)	0.856 (prevalence)	win, sharpens in-flight
Account takeover at origination	0.029 ISO-only -> 1.000 fused with pain.001 channel context	ISO fields alone	win, multi-source
Velocity burst (dense fleet)	PR-AUC 0.822	0.222 order-blind	win, timing-only
Next payment: days until (held-out customers)	MAE 23.5 d	25.9 / 32.5 (median / last gap)	win, modest
Next payment: occurrence within 35 d	PR-AUC 0.579	tree 0.655, rule 0.637	loss - trees win
Next payment: amount	MAE 33.0k	21.8k (history median)	loss
Rail routing	acc 0.556	tree 0.645, majority 0.478	loss - C2 territory (SWIFT F1 = 1.00)
Terminal status at intake	acc 0.279	majority 0.740	loss - C2 territory
Reject reason (ISO code)	acc 0.102	'none' dominates	loss
Cancel / return likelihood at intake	PR-AUC at prevalence (0.030 / 0.018)	-	null - event-driven, not intent-visible
Message-order signal (tracker outcome)	sequence macro-F1 0.270 vs pooled 0.275 (-0.5 pp)	bag of messages	null - prefixes are 2-3 near-fixed-order messages

Table 2: every exposed prediction on the lifecycle corpus, H200 full runs. The losses and nulls are load-bearing: they route those fields to trees, rules, or nothing.

Two rows deserve prose. The account-takeover result is the multi-source argument in one line: ISO clearing fields carry no trace of a hijacked session (0.029, chance at 2.7% prevalence), while the same frozen encoder reading the pain.001 origination context alongside them separates the classes completely - on synthetic data whose generator plants exactly that correlation, so the number demonstrates the fusion path works, not that real fraud is this easy. The message-order null is the opposite lesson. A single payment's pre-outcome trail is two or three messages in near-fixed order; there is nothing for a sequence model to read that a bag of messages does not already say, and the measured -0.5 points says exactly that. Temporal signal in payments lives at the entity level (C3, velocity), not inside one payment's message chain.

7 Serving

The deployed artifact is the frozen encoder, the frozen history encoder, and a set of linear heads with per-head isotonic calibration fitted on held-out prefixes - roughly 260MB, CPU-servable. Scoring is stateless: the engine sends a payment row, or a message prefix, or an account's recent history, and receives calibrated probabilities plus a predicted-lifecycle object (settlement outcome distribution, remaining ETA, cancel and return probabilities). Uncalibrated heads had produced scores like 0.94 for a 3%-prevalence event; calibration restored readability (the same head now emits 0.19). Explanations come from column occlusion against the frozen encoder - occlude a field, re-embed, measure the shift - which is faithful to the model by construction and costs about twenty embeddings per explained row. Two operational caveats are measured rather than suspected: brand-new accounts degrade routing confidence to near-uniform (identity features carry most of the rail signal), and a bundle trained under scikit-learn 1.9 requires a one-line attribute shim to load under 1.7.

8 Limitations

Everything here is synthetic, and the generators were written by the same team that wrote the models. We defended against self-deception where we knew how - matched aggregates for C3/C4, prevalence-honest PR-AUC, held-out actors, visibility-filtered prefixes, thresholds committed in advance - but a generator cannot invent the correlations of real payment traffic, and the ATO fusion number in particular measures a planted signal. Each full-scale number is one seeded run; we report no variance across seeds. C5 is measured on one task family with one 1.3B LLM, and says nothing about text-emitting or few-shot settings. The rare-event heads (cancel at 3.0%, return at 1.8% of 20,000 payments) sit at prevalence and would need two orders of magnitude more data before failure means anything. The published v1 checkpoint predates the lifecycle work and scores only the original task suite. Real transaction data, multiple seeds, and an entity-level generator whose successive payments correlate are the three shortest paths to making these numbers mean more.

9 Conclusion

PULSE's frozen encoder does carry many payment decisions - provided one is honest about which ones. Where the signal is a rule over a complete row, a gradient-boosted tree wins and should be shipped instead (C2b, routing, status). Where the signal is order, timing, partial information, or a second source the tree never sees, the frozen representation wins by margins of 38 to 60 PR-AUC points (C3, C4, velocity, in-flight, fusion). The frozen LLM, the most expensive component the source architecture allows, contributes 1.3 points and was removed from serving (C5). The claims ledger, generators, and 197 tests reproduce every number in this paper from two commands.

References

- [1] Raman, B., Ganesh, S., Veloso, M. (2024). Scalable Representation Learning for Multimodal Tabular Transactions. arXiv:2410.07851. NeurIPS 2024 Table Representation Learning workshop.
- [2] Hermans, A., Beyer, L., Leibe, B. (2017). In Defense of the Triplet Loss for Person Re-Identification. arXiv:1703.07737.
- [3] Li, X. L., Liang, P. (2021). Prefix-Tuning: Optimizing Continuous Prompts for Generation. ACL. arXiv:2101.00190.
- [4] Lester, B., Al-Rfou, R., Constant, N. (2021). The Power of Scale for Parameter-Efficient Prompt Tuning. EMNLP. arXiv:2104.08691.
- [5] Li, Y., et al. (2023). Textbooks Are All You Need II: phi-1.5 Technical Report. arXiv:2309.05463.
- [6] Prokhorenkova, L., et al. (2018). CatBoost: Unbiased Boosting with Categorical Features. NeurIPS.
- [7] Padhi, I., et al. (2021). Tabular Transformers for Modeling Multivariate Time Series (TabBERT). ICASSP. arXiv:2011.01843.
- [8] Babaev, D., et al. (2022). CoLES: Contrastive Learning for Event Sequences with Self-Supervision. SIGMOD. arXiv:2002.08232.
- [9] Zhang, D., et al. (2023). FATA-Trans: Field And Time-Aware Transformer for Sequential Tabular Data. CIKM. arXiv:2310.13818.
- [10] Visa Research (2025). TransactionGPT. arXiv:2511.08939.
- [11] Revolut (2026). PRAGMA: Revolut Foundation Model. arXiv:2604.08649.
- [12] Hollmann, N., et al. (2025). Accurate Predictions on Small Data with a Tabular Foundation Model (TabPFN). Nature 637.
- [13] Zadrozny, B., Elkan, C. (2002). Transforming Classifier Scores into Accurate Multiclass Probability Estimates. KDD.